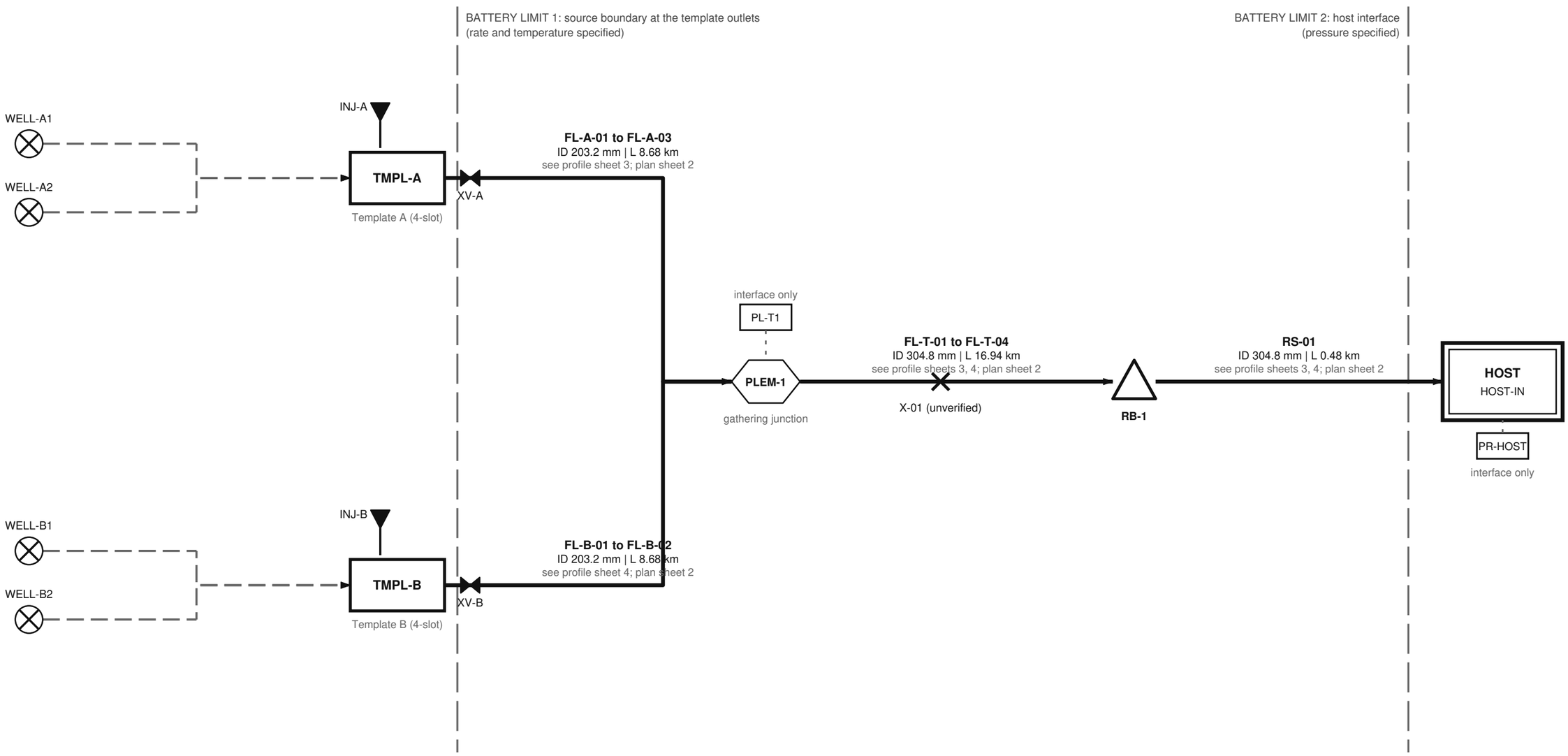
FA-DEMO-001-DWG-01 Field and network schematic (connectivity)



SCREENING - synthetic inputs; not for design or construction

LEGEND

- Modelled hydraulic run (flow direction by arrow)
- Non-modelled jumper (aggregated)
- Battery limit (model boundary)
- Template / host (box symbols)
- MEG injection point (modelled)
- Isolation valve (line-up variants)
- Crossing (verification status stated)
- Pig launcher/receiver: interface only, no simulation

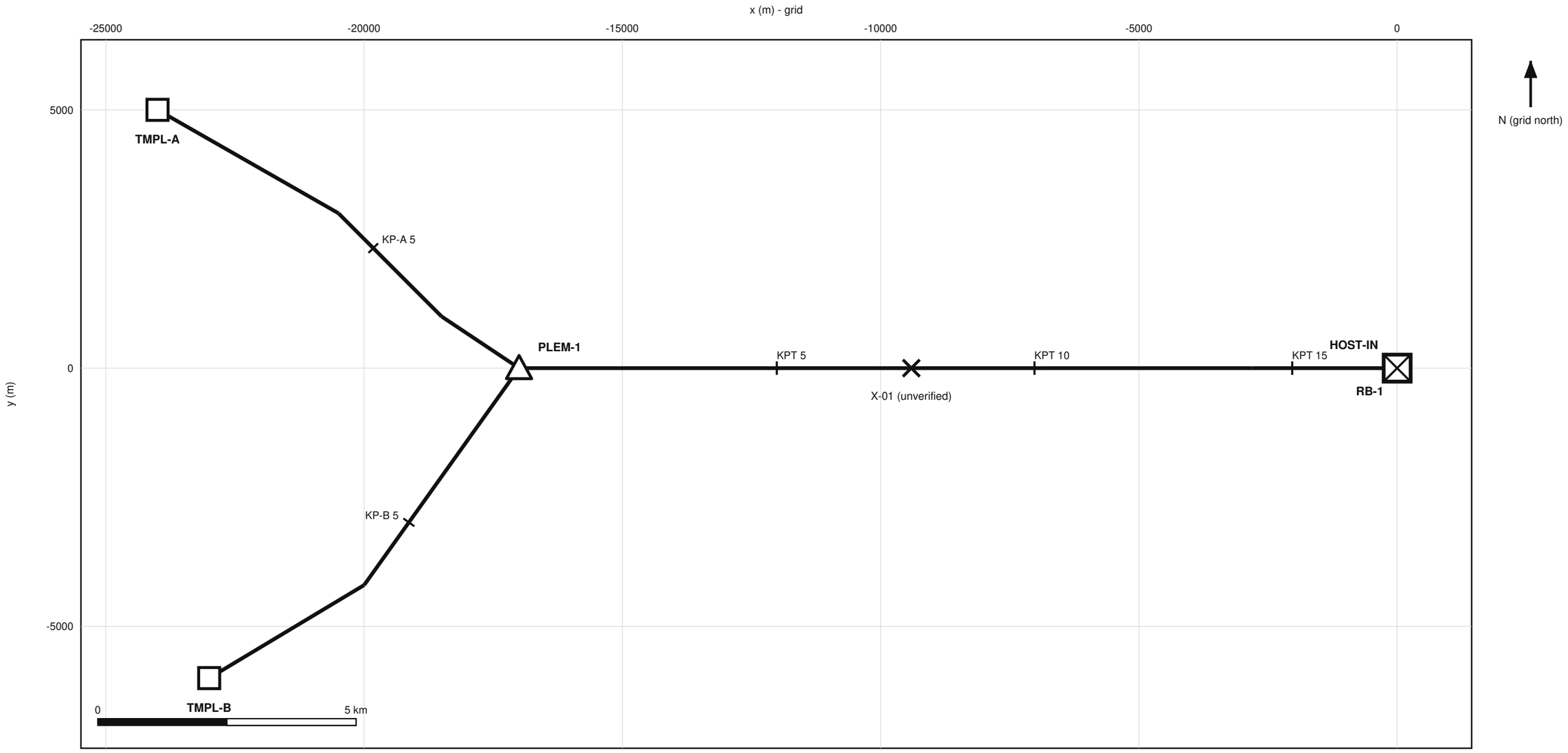
GENERAL NOTES

1. Connectivity schematic only: positions and lengths on the sheet do not represent geography or scale (plan: sheet 2).
2. Line tags are the stable segment IDs of the route file; waypoints between assets are collapsed into one run (all segment IDs are listed on the run).
3. Solid lines are modelled hydraulic segments; dashed grey jumpers are not modelled (stream aggregated at the template).
4. Topology variants in the case matrix: B-ISOLATED, DEFAULT. Isolation by XV-A/XV-B keeps all asset and segment IDs.
5. Battery limits are modelling boundaries, not facility or licence boundaries.
6. Issue purpose: DEMONSTRATION - SYNTHETIC DATA; readiness SCREENING (assigned by the toolkit gates, not by a person).
7. Run reference: dataset bd5858a7e40e53e449cc4021, model fa_solver steady network model, NeqSim neqsim-3.23.0-java25.jar.
8. Units: pressure bara, temperature degC, lengths m, rates kg/s, time h.

Rev	Date	Description
A	2026-10-25	First issue of the synthetic demonstration package

Company: NeqSim demonstration (no company)	Project: Synthetic tie-back (no real field)
Title: **Synthetic Offshore Tie-back - Flow-Assurance Study (Demonstration) - Field and network schematic (connectivity)**	
Drawing number: **FA-DEMO-001-DWG-01**	Revision: A / Status: **SCREENING**
Sheet: Sheet 1 of 10 / Date: 2026-10-25	Scale: NOT TO SCALE / Format: A3 landscape
Drawn by: not assigned	Checked by: not assigned / Approved by: not assigned

FA-DEMO-001-DWG-02 Pipeline route plan



SYNTHETIC LOCAL GRID - NOT GEOREFERENCED

SCREENING - synthetic inputs; not for design or construction

LEGEND

- Modelled hydraulic route (plan)
- Non-modelled jumper
- Template / host interface
- Gathering PLEM
- Riser base
- Crossing (unverified)
- Ticks: KP marks every 5 km of chainage

GENERAL NOTES

1. Coordinate reference: SYNTHETIC-LOCAL-XY (metres, x east, y north); not a geodetic CRS. Horizontal datum: none (synthetic local grid). Vertical datum: MSL (synthetic); elevations positive up, seabed negative.
2. Plan scale 1:81600 at A3 (1 mm = 81.6 m). KP marks every 5 km of measured chainage, interpolated along the plan polyline in proportion to measured length.
3. Chainage convention: Template outlet of each source (path start); chainage runs along the flow direction
4. KP-A, KP-B: branch runs from the template outlets; KPT: trunk from the gathering PLEM. Jumpers (dashed) are not modelled hydraulically.
5. Safety zones, third-party infrastructure and seabed hazards were not supplied and are not drawn; only the crossing marked is shown and it is unverified.
6. Pig interfaces and injection points have no plan coordinates and are shown on the connectivity schematic (sheet 1) only.
7. Issue purpose: DEMONSTRATION - SYNTHETIC DATA; readiness SCREENING (assigned by the toolkit gates, not by a person).
8. Run reference: dataset bd5858a7e40e53e449cc4021, model fa_solver steady network model, NeqSim neqsim-3.23.0-java25.jar.
9. Units: pressure bara, temperature degC, lengths m, rates kg/s, time h.

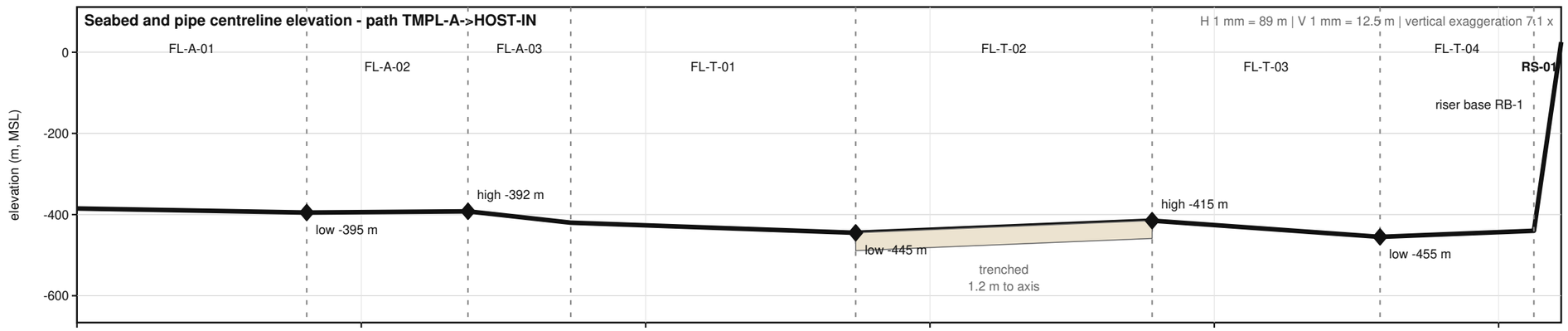
Rev	Date	Description
A	2026-10-25	First issue of the synthetic demonstration package

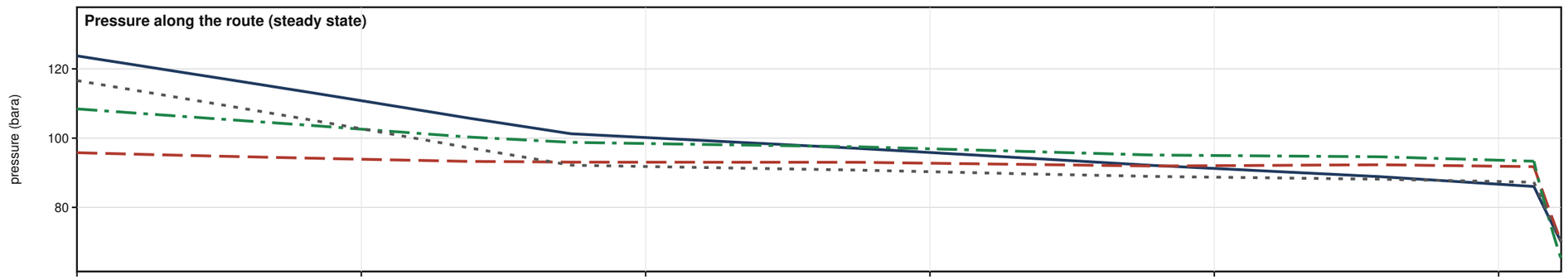
Company: NeqSim demonstration (no company)	Project: Synthetic tie-back (no real field)

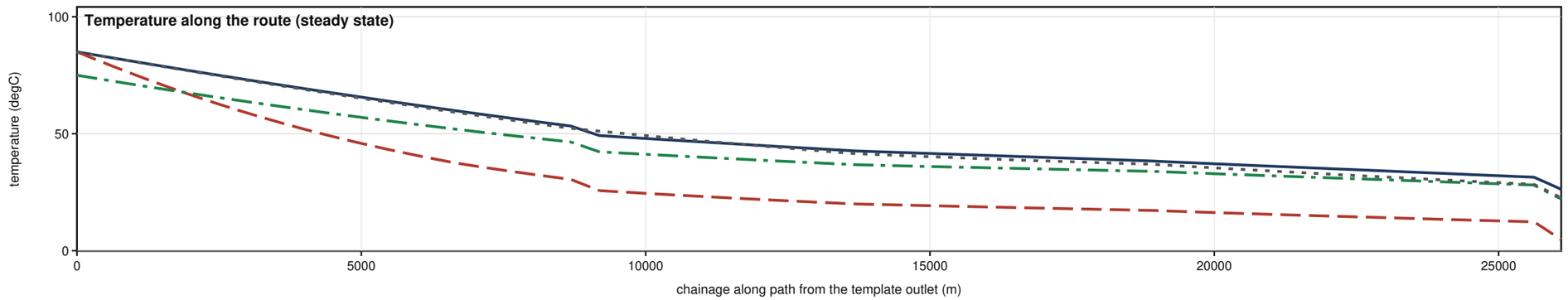
Title: **Synthetic Offshore Tie-back - Flow-Assurance Study (Demonstration) - Pipeline route plan**

Drawing number: **FA-DEMO-001-DWG-02**	Revision: A	Status: **SCREENING**	
Sheet: Sheet 2 of 10	Date: 2026-10-25	Scale: 1:81600 (horizontal, A3)	Format: A3 landscape
Drawn by: not assigned	Checked by: not assigned	Approved by: not assigned	

FA-DEMO-001-DWG-03 Longitudinal profile with pressure and temperature - path TMPL-A->HOST-IN







SCREENING - synthetic inputs; not for design or construction

LEGEND

- BASE - Base - plateau, early life
- TURNDOWN - Turndown - minimum stable rate
- LATE - Late life - high water, declining rate
- ALT-B-ISOLATED - Alternate line-up - template B isolated
- Interior low / high point of the seabed profile

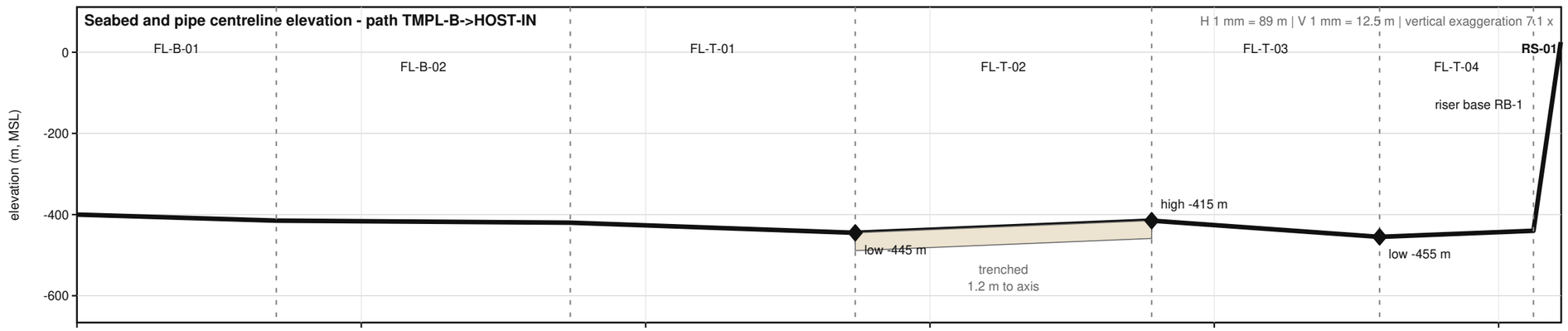
GENERAL NOTES

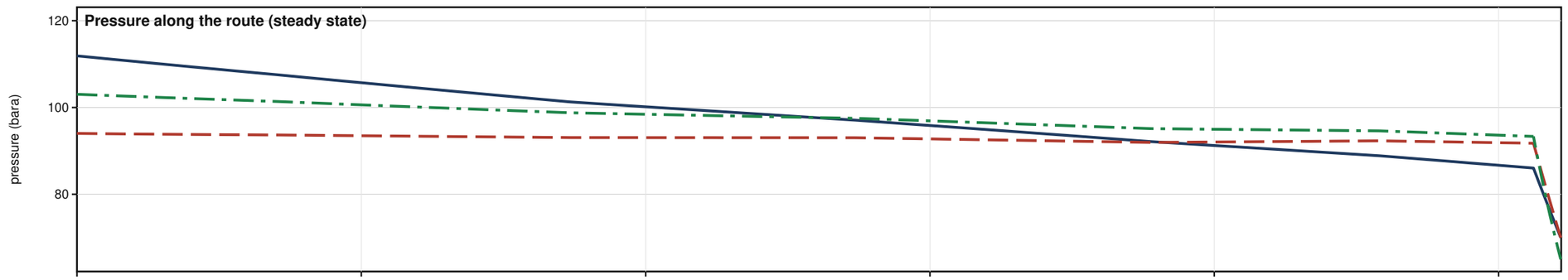
1. Path TMPL-A->HOST-IN (source to host interface). Chainage is measured (as-laid) route length from the template outlet, along the flow direction; horizontal projection and elevation are separate quantities.
2. Pressure and temperature are steady-state Beggs and Brill results (fa_solver steady network model); the three panels share the same chainage axis.
3. Elevation is the seabed/pipe centreline at the route nodes joined by straight lines; elevations between nodes are not supplied. Burial is shown where the route file states it.
4. Cases plotted: design and scenario cases whose line-up includes this path; mitigation cases are on sheets 7-8.
5. The host-interface pressure is the specified boundary; source pressures are calculated (backward solve).
6. Issue purpose: DEMONSTRATION - SYNTHETIC DATA; readiness SCREENING (assigned by the toolkit gates, not by a person).
7. Run reference: dataset bd5858a7e40e53e449cc4021, model fa_solver steady network model, NeqSim neqsim-3.23.0-java25.jar.
8. Units: pressure bara, temperature degC, lengths m, rates kg/s, time h.

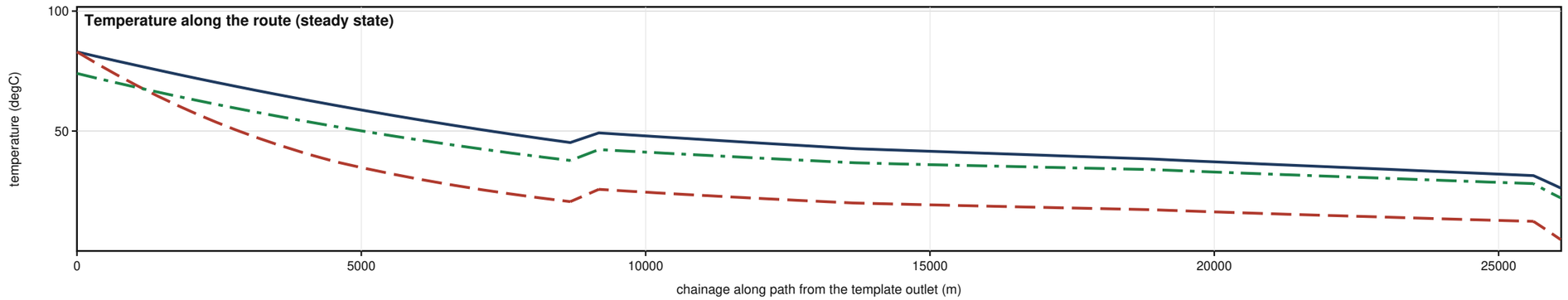
Rev	Date	Description
A	2026-10-25	First issue of the synthetic demonstration package

Company: NeqSim demonstration (no company)	Project: Synthetic tie-back (no real field)
Title: **Synthetic Offshore Tie-back - Flow-Assurance Study (Demonstration) - Longitudinal profile with pressure and temperature - path TMPL-A->HOST-IN**	
Drawing number: **FA-DEMO-001-DWG-03**	Revision: A / Status: **SCREENING**
Sheet: Sheet 3 of 10 / Date: 2026-10-25	Scale: H 1:89400 V 1:12540 / Format: A3 landscape
Drawn by: not assigned	Checked by: not assigned / Approved by: not assigned

FA-DEMO-001-DWG-04 Longitudinal profile with pressure and temperature - path TMPL-B->HOST-IN







SCREENING - synthetic inputs; not for design or construction

LEGEND

- BASE - Base - plateau, early life
- TURNDOWN - Turndown - minimum stable rate
- LATE - Late life - high water, declining rate
- Interior low / high point of the seabed profile

GENERAL NOTES

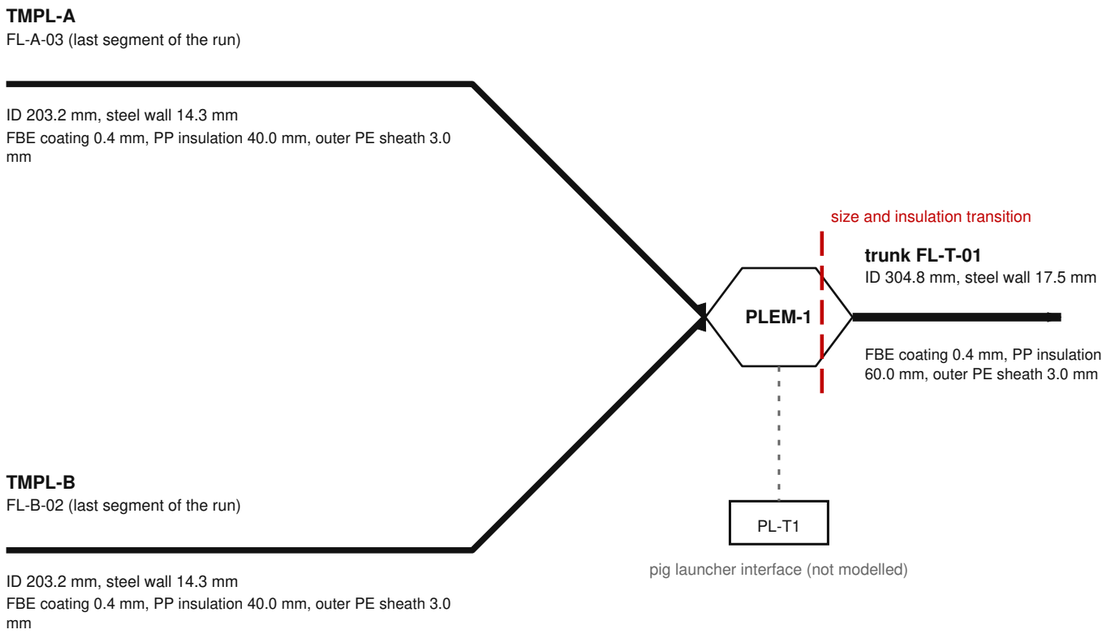
1. Path TMPL-B->HOST-IN (source to host interface). Chainage is measured (as-laid) route length from the template outlet, along the flow direction; horizontal projection and elevation are separate quantities.
2. Pressure and temperature are steady-state Beggs and Brill results (fa_solver steady network model); the three panels share the same chainage axis.
3. Elevation is the seabed/pipe centreline at the route nodes joined by straight lines; elevations between nodes are not supplied. Burial is shown where the route file states it.
4. Cases plotted: design and scenario cases whose line-up includes this path; mitigation cases are on sheets 7-8.
5. The host-interface pressure is the specified boundary; source pressures are calculated (backward solve).
6. Issue purpose: DEMONSTRATION - SYNTHETIC DATA; readiness SCREENING (assigned by the toolkit gates, not by a person).
7. Run reference: dataset bd5858a7e40e53e449cc4021, model fa_solver steady network model, NeqSim neqsim-3.23.0-java25.jar.
8. Units: pressure bara, temperature degC, lengths m, rates kg/s, time h.

Rev	Date	Description
A	2026-10-25	First issue of the synthetic demonstration package

Company: NeqSim demonstration (no company)		Project: Synthetic tie-back (no real field)	
Title: **Synthetic Offshore Tie-back - Flow-Assurance Study (Demonstration) - Longitudinal profile with pressure and temperature - path TMPL-B->HOST-IN**			
Drawing number: **FA-DEMO-001-DWG-04**		Revision: A	Status: **SCREENING**
Sheet: Sheet 4 of 10	Date: 2026-10-25	Scale: H 1:89400 V 1:12540	Format: A3 landscape
Drawn by: not assigned	Checked by: not assigned	Approved by: not assigned	

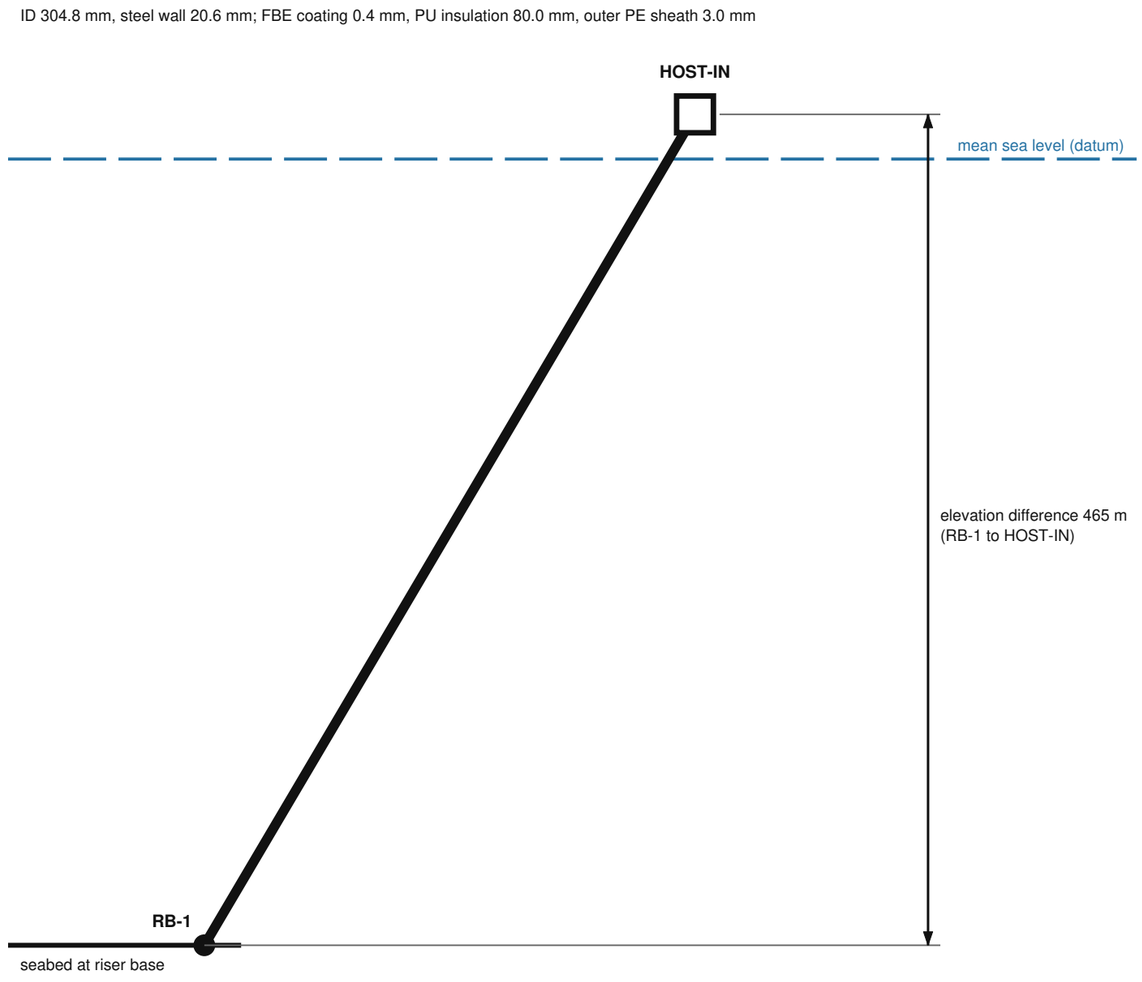
FA-DEMO-001-DWG-05 Riser arrangement and PLEM tie-in detail (schematic)

PLEM-1 tie-in - plan schematic, NOT TO SCALE



Hydraulic tie-in only: structural, fabrication, weld and support design are outside this study; no dimensions are implied.

Riser - elevation schematic, NOT TO SCALE



measured length 480 m

plan distance of the end nodes 100 m

Geometry between the end nodes is not supplied: the riser is drawn as a straight chord, bends, supports and hang-off are not shown.

SCREENING - synthetic inputs; not for design or construction

LEGEND

- Pipe run
- Insulation / size transition
- Mean sea level (datum)
- Interface only (not modelled)

GENERAL NOTES

1. Schematic arrangement: shows identifiers, nominal sizes, supplied elevations and layer thicknesses only.
2. No fabrication dimensions, welds, supports, bend radii or hang-off details are shown or implied; ISO 129-1 dimension presentation is not applied (see standards register).
3. Insulation and size transitions follow the route file; a transition at a node is reported when the thermal build-up of adjacent segments differs.
4. Elevations are relative to the stated vertical datum (positive up).
5. Issue purpose: DEMONSTRATION - SYNTHETIC DATA; readiness SCREENING (assigned by the toolkit gates, not by a person).
6. Run reference: dataset bd5858a7e40e53e449cc4021, model fa_solver steady network model, NeqSim neqsim-3.23.0-java25.jar.
7. Units: pressure bara, temperature degC, lengths m, rates kg/s, time h.

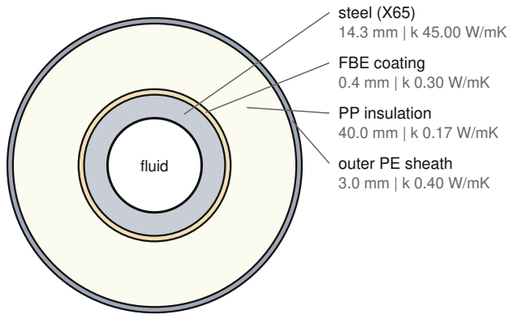
Rev	Date	Description
A	2026-10-25	First issue of the synthetic demonstration package

Company: NeqSim demonstration (no company)		Project: Synthetic tie-back (no real field)	
Title: **Synthetic Offshore Tie-back - Flow-Assurance Study (Demonstration) - Riser arrangement and PLEM tie-in detail (schematic)**			
Drawing number: **FA-DEMO-001-DWG-05**		Revision: A	Status: **SCREENING**
Sheet: Sheet 5 of 10	Date: 2026-10-25	Scale: NOT TO SCALE	Format: A3 landscape
Drawn by: not assigned	Checked by: not assigned	Approved by: not assigned	

FA-DEMO-001-DWG-06 Pipe and thermal cross-sections

Section A: FL-A-01 ... FL-B-02

applies to 5 segments



ID 203.2 mm OD 318.6 mm

U (inner-area basis) 5.43 W/m2K

U (outer-area basis) 3.46 W/m2K

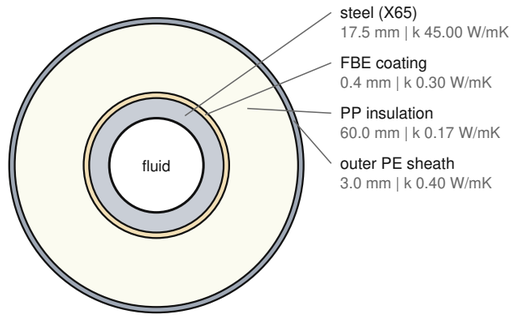
external boundary: external film

h_ext = 500 W/m2K (input)

NOT TO SCALE (thin layers drawn thicker)

Section B: FL-T-01 ... FL-T-04

applies to 3 segments



ID 304.8 mm OD 466.6 mm

U (inner-area basis) 3.59 W/m2K

U (outer-area basis) 2.35 W/m2K

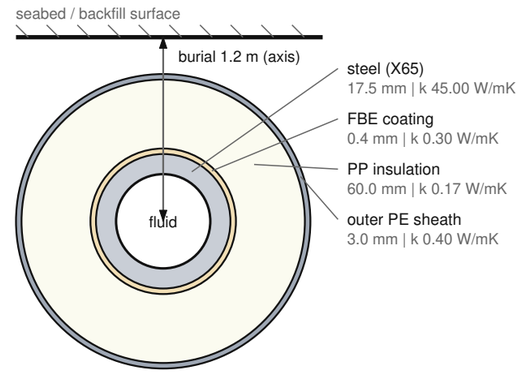
external boundary: external film

h_ext = 500 W/m2K (input)

NOT TO SCALE (thin layers drawn thicker)

Section C: FL-T-02

applies to 1 segment



ID 304.8 mm OD 466.6 mm

U (inner-area basis) 1.71 W/m2K

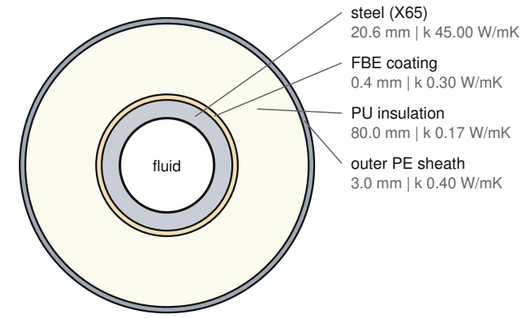
U (outer-area basis) 1.12 W/m2K

external boundary: soil shape factor

NOT TO SCALE (thin layers drawn thicker)

Section D: RS-01

applies to 1 segment



ID 304.8 mm OD 512.8 mm

U (inner-area basis) 2.87 W/m2K

U (outer-area basis) 1.71 W/m2K

external boundary: external film

h_ext = 300 W/m2K (input)

NOT TO SCALE (thin layers drawn thicker)

SCREENING - synthetic inputs; not for design or construction

LEGEND

Concentric layers from the fluid outward: steel wall, coating, insulation, sheath

Leader line to the layer table

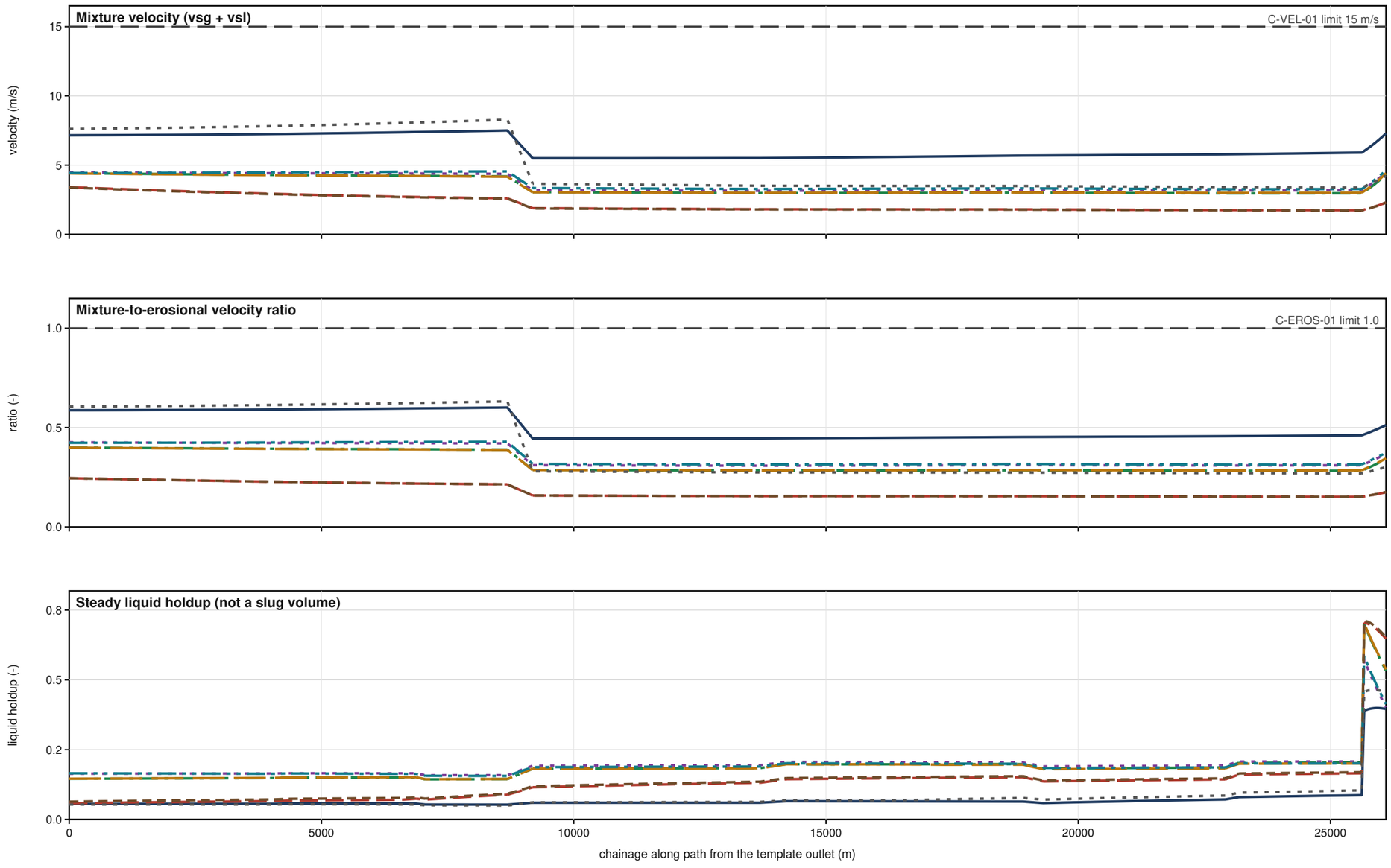
GENERAL NOTES

1. U-values are referenced to the INNER pipe surface (pi x ID x L), the basis used by the hydraulic solver (SPECIFIED_U); the outer-area value is shown for comparison only.
2. Internal film resistance is neglected: U is larger than the true value, which is conservative for arrival temperature and cooldown.
3. External boundary: seawater film coefficient (input) for exposed segments, or the buried-pipe shape factor with the stated soil conductivity.
4. Layer thermal properties are inputs of the route file; wet-insulation degradation, field joints and end effects are not modelled.
5. Sections are schematic and not to scale; they show no fabrication detail.
6. Issue purpose: DEMONSTRATION - SYNTHETIC DATA; readiness SCREENING (assigned by the toolkit gates, not by a person).
7. Run reference: dataset bd5858a7e40e53e449cc4021, model fa_solver steady network model, NeqSim neqsim-3.23.0-java25.jar.
8. Units: pressure bara, temperature degC, lengths m, rates kg/s, time h.

Rev	Date	Description
A	2026-10-25	First issue of the synthetic demonstration package

Company: NeqSim demonstration (no company)	Project: Synthetic tie-back (no real field)
Title: **Synthetic Offshore Tie-back - Flow-Assurance Study (Demonstration) - Pipe and thermal cross-sections**	
Drawing number: **FA-DEMO-001-DWG-06**	Revision: A / Status: **SCREENING**
Sheet: Sheet 6 of 10 / Date: 2026-10-25	Scale: NOT TO SCALE / Format: A3 landscape
Drawn by: not assigned	Checked by: not assigned / Approved by: not assigned

FA-DEMO-001-DWG-07 Velocity, erosional ratio and liquid holdup along the route



Critical sections (any path)

BASE
v_mix max 7.5 m/s at KP 8.7, FL-A-03
ratio max 0.6 at KP 8.7, FL-A-03
holdup max 0.4 at KP 25.9, RS-01

TURNDOWN
v_mix max 3.4 m/s at KP 0.0, FL-A-01
ratio max 0.2 at KP 0.0, FL-A-01
holdup max 0.7 at KP 25.7, RS-01

LATE
v_mix max 4.4 m/s at KP 0.0, FL-A-01
ratio max 0.4 at KP 0.0, FL-A-01
holdup max 0.7 at KP 25.7, RS-01

LATE-M1
v_mix max 4.5 m/s at KP 0.0, FL-A-01
ratio max 0.4 at KP 0.0, FL-A-01
holdup max 0.6 at KP 25.7, RS-01

LATE-M2
v_mix max 4.4 m/s at KP 26.1, RS-01
ratio max 0.4 at KP 0.0, FL-A-01
holdup max 0.7 at KP 25.7, RS-01

LATE-M3
v_mix max 4.6 m/s at KP 26.1, RS-01
ratio max 0.4 at KP 8.7, FL-A-03
holdup max 0.6 at KP 25.7, RS-01

TURN-M4
v_mix max 3.4 m/s at KP 0.0, FL-A-01
ratio max 0.2 at KP 0.0, FL-A-01
holdup max 0.7 at KP 25.7, RS-01

ALT-B-ISOLATED
v_mix max 8.3 m/s at KP 8.7, FL-A-03
ratio max 0.6 at KP 8.7, FL-A-03
holdup max 0.5 at KP 25.9, RS-01

SCREENING - synthetic inputs; not for design or construction

LEGEND

- BASE - Base - plateau, early life
- TURNDOWN - Turndown - minimum stable rate
- LATE - Late life - high water, declining rate
- LATE-M1 - Late life - MEG dose raised to 30 wt % aqueous
- LATE-M2 - Late life - trunk insulation +30 mm, baseline MEG dose
- LATE-M3 - Late life - MEG 30 wt % and branch insulation 80 mm
- TURN-M4 - Turndown - MEG dose raised to 40 wt % aqueous
- ALT-B-ISOLATED - Alternate line-up - template B isolated
- Project limit (sourced criterion)

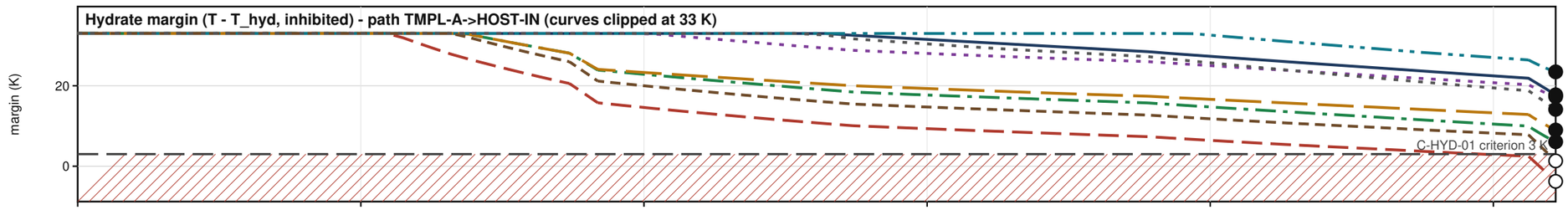
GENERAL NOTES

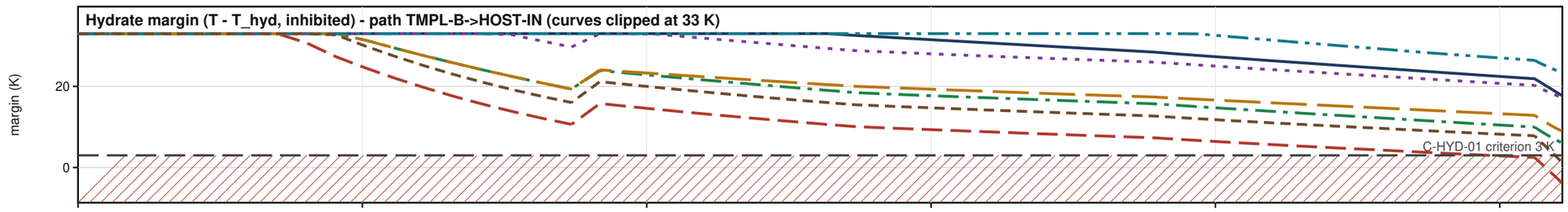
1. Path TMPL-A->HOST-IN shown; downstream of the PLEM the other path is identical, branch values of the other path are in the dataset.
2. Velocity limits are project criteria (basis, acceptance_criteria); the erosional velocity uses the project C-factor and is not a DNV-RP-O501 assessment.
3. Holdup is the steady Beggs and Brill value; it must not be read as transient slug volume or as a liquid-handling capacity check.
4. Critical-section column lists the maximum over all active paths of each case (KP = chainage in km along the stated path).
5. Issue purpose: DEMONSTRATION - SYNTHETIC DATA; readiness SCREENING (assigned by the toolkit gates, not by a person).
6. Run reference: dataset bd5858a7e40e53e449cc4021, model fa_solver steady network model, NeqSim neqsim-3.23.0-java25.jar.
7. Units: pressure bara, temperature degC, lengths m, rates kg/s, time h.

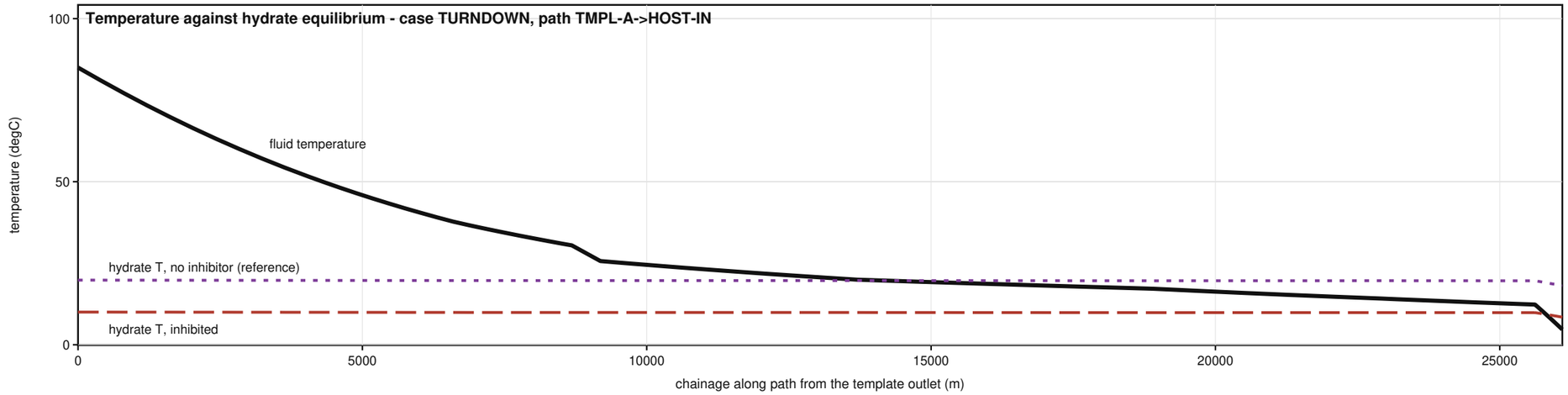
Rev	Date	Description
A	2026-10-25	First issue of the synthetic demonstration package

Company: NeqSim demonstration (no company)	Project: Synthetic tie-back (no real field)
Title: **Synthetic Offshore Tie-back - Flow-Assurance Study (Demonstration) - Velocity, erosional ratio and liquid holdup along the route**	
Drawing number: **FA-DEMO-001-DWG-07**	Revision: A / Status: **SCREENING**
Sheet: Sheet 7 of 10	Date: 2026-10-25 / Scale: NOT TO SCALE / Format: A3 landscape
Drawn by: not assigned	Checked by: not assigned / Approved by: not assigned

FA-DEMO-001-DWG-08 Hydrate margin and critical sections







Critical sections and criterion

BASE
min margin 17.7 K at KP 26.1, RS-01
C-HYD-01 PASS (limit 3 K)

TURNDOWN
min margin -3.8 K at KP 26.1, RS-01
C-HYD-01 FAIL (limit 3 K)

LATE
min margin 6.1 K at KP 26.1, RS-01
C-HYD-01 PASS (limit 3 K)

LATE-M1
min margin 17.3 K at KP 26.1, RS-01
C-HYD-01 PASS (limit 3 K)

LATE-M2
min margin 9.0 K at KP 26.1, RS-01
C-HYD-01 PASS (limit 3 K)

LATE-M3
min margin 23.5 K at KP 26.1, RS-01
C-HYD-01 PASS (limit 3 K)

TURN-M4
min margin 1.3 K at KP 26.1, RS-01
C-HYD-01 FAIL (limit 3 K)

ALT-B-ISOLATED
min margin 14.1 K at KP 26.1, RS-01
C-HYD-01 PASS (limit 3 K)

SCREENING - synthetic inputs; not for design or construction

LEGEND

- BASE - Base - plateau, early life
- TURNDOWN - Turndown - minimum stable rate
- LATE - Late life - high water, declining rate
- LATE-M1 - Late life - MEG dose raised to 30 wt % aqueous
- LATE-M2 - Late life - trunk insulation +30 mm, baseline MEG dose
- LATE-M3 - Late life - MEG 30 wt % and branch insulation 80 mm
- TURN-M4 - Turndown - MEG dose raised to 40 wt % aqueous
- ALT-B-ISOLATED - Alternate line-up - template B isolated
- Project hydrate-margin criterion (hatched = below)
- Minimum margin, criterion PASS (filled)
- Minimum margin, criterion FAIL (hollow)

GENERAL NOTES

1. Margin = fluid temperature minus the CPA hydrate equilibrium temperature at the local pressure and the aqueous inhibitor mass fraction of the local stream (equilibrium, not plugging time).
2. The inhibitor mass fraction follows the injection rate balance; losses to gas and condensate are not modelled (upper bound on the concentration).
3. The 3 K margin is a project criterion (source: DEMO-DB-001 (synthetic project criterion; not a code requirement)), not a code value. The hatched band is the region below the criterion.
4. Bottom panel: governing case TURNDOWN; line styles are labelled on the panel and do not follow the case colours of the legend; the uninhibited curve is a reference, not an operating mode.
5. Hydrate temperatures are interpolated in ln(p) on a pressure grid; the interpolation check is reported in the validation appendix.
6. Issue purpose: DEMONSTRATION - SYNTHETIC DATA; readiness SCREENING (assigned by the toolkit gates, not by a person).
7. Run reference: dataset bd5858a7e40e53e449cc4021, model fa_solver steady network model, NeqSim neqsim-3.23.0-java25.jar.
8. Units: pressure bara, temperature degC, lengths m, rates kg/s, time h.

Rev	Date	Description
A	2026-10-25	First issue of the synthetic demonstration package

Company: NeqSim demonstration (no company)	Project: Synthetic tie-back (no real field)
Title: **Synthetic Offshore Tie-back - Flow-Assurance Study (Demonstration) - Hydrate margin and critical sections**	
Drawing number: **FA-DEMO-001-DWG-08**	Revision: A; Status: **SCREENING**
Sheet: Sheet 8 of 10	Date: 2026-10-25; Scale: NOT TO SCALE; Format: A3 landscape
Drawn by: not assigned	Checked by: not assigned; Approved by: not assigned

FA-DEMO-001-DWG-09 Operating-limit summary by case

Criterion (limit)	BASE design	TURNDOWN scenario	LATE scenario	LATE-M1 mitigation	LATE-M2 mitigation	LATE-M3 mitigation	TURN-M4 mitigation	ALT-B-ISOLATED scenario
C-HYD-01 hydrate margin min >= 3 K **overall: FAIL**	17.7 ● **PASS**	-3.8 △ **FAIL**	6.1 ● **PASS**	17.3 ● **PASS**	9.0 ● **PASS**	23.5 ● **PASS**	1.3 △ **FAIL**	14.1 ● **PASS**
C-SRC-01 source pressure max <= 180 bara **overall: PASS**	123.8 ● **PASS**	95.8 ● **PASS**	108.4 ● **PASS**	108.2 ● **PASS**	108.8 ● **PASS**	109.6 ● **PASS**	96.7 ● **PASS**	116.6 ● **PASS**
C-VEL-01 mixture velocity max <= 15 m/s **overall: PASS**	7.5 ● **PASS**	3.4 ● **PASS**	4.4 ● **PASS**	4.5 ● **PASS**	4.4 ● **PASS**	4.6 ● **PASS**	3.4 ● **PASS**	8.3 ● **PASS**
C-EROS-01 erosional ratio max <= 1 **overall: PASS**	0.6 ● **PASS**	0.2 ● **PASS**	0.4 ● **PASS**	0.4 ● **PASS**	0.4 ● **PASS**	0.4 ● **PASS**	0.2 ● **PASS**	0.6 ● **PASS**
C-COOL-01 cooldown time min >= 8 h **overall: FAIL**	6.6 △ **FAIL**	0.0 △ **FAIL**	2.7 △ **FAIL**	7.0 △ **FAIL**	3.4 △ **FAIL**	12.9 ● **PASS**	0.0 △ **FAIL**	not applicable

Threat applicability matrix (status of every threat; an open or unsupported threat is never a safe result)

● **T-HYD-STEADY CALCULATED**	Steady pressure, temperature, phase rates, velocity and holdup profiles	✕ **T-SEVERE-SLUG UNSUPPORTED**	Severe slugging and terrain-induced slugging
● **T-CAP CALCULATED**	Capacity, required source pressure (backpressure) and turndown envelope	△ **T-WAX OPEN**	Wax appearance temperature and deposition
● **T-HYDRATE CALCULATED**	Hydrate equilibrium margin, inhibited and uninhibited, inhibitor injection balance	□ **T-ASPHALTENE NOT APPLICABLE**	Asphaltene stability
● **T-THERMAL CALCULATED**	Heat loss, arrival temperature and insulation alternatives	△ **T-SCALE OPEN**	Mineral scale and brine compatibility
● **T-COOLDOWN CALCULATED**	Cooldown time to the hydrate equilibrium temperature after shut-in	△ **T-CORROSION OPEN**	CO2 corrosion (NORSOK M-506 basis) and inspected metal loss
● **T-EROSION-VEL CALCULATED**	Mixture velocity against the project erosional-velocity limit	□ **T-SAND NOT APPLICABLE**	Sand production and erosion
✕ **T-STARTUP UNSUPPORTED**	Startup, restart, ramp-up, depressurisation, liquid accumulation and sweep-out	□ **T-SURGE NOT APPLICABLE**	Acoustic surge and water/liquid hammer
✕ **T-PIGGING UNSUPPORTED**	Pigging and slug-receiving requirements	△ **T-INJ-PHASE OPEN**	Injection-fluid phase behaviour (inhibitor partitioning, freezing, losses)

SCREENING - synthetic inputs; not for design or construction

LEGEND

- ● PASS / calculated
- △ FAIL / open
- □ Not applicable (reason stated)
- ✕ Unsupported (external hand-off)

Status is always also written as a word

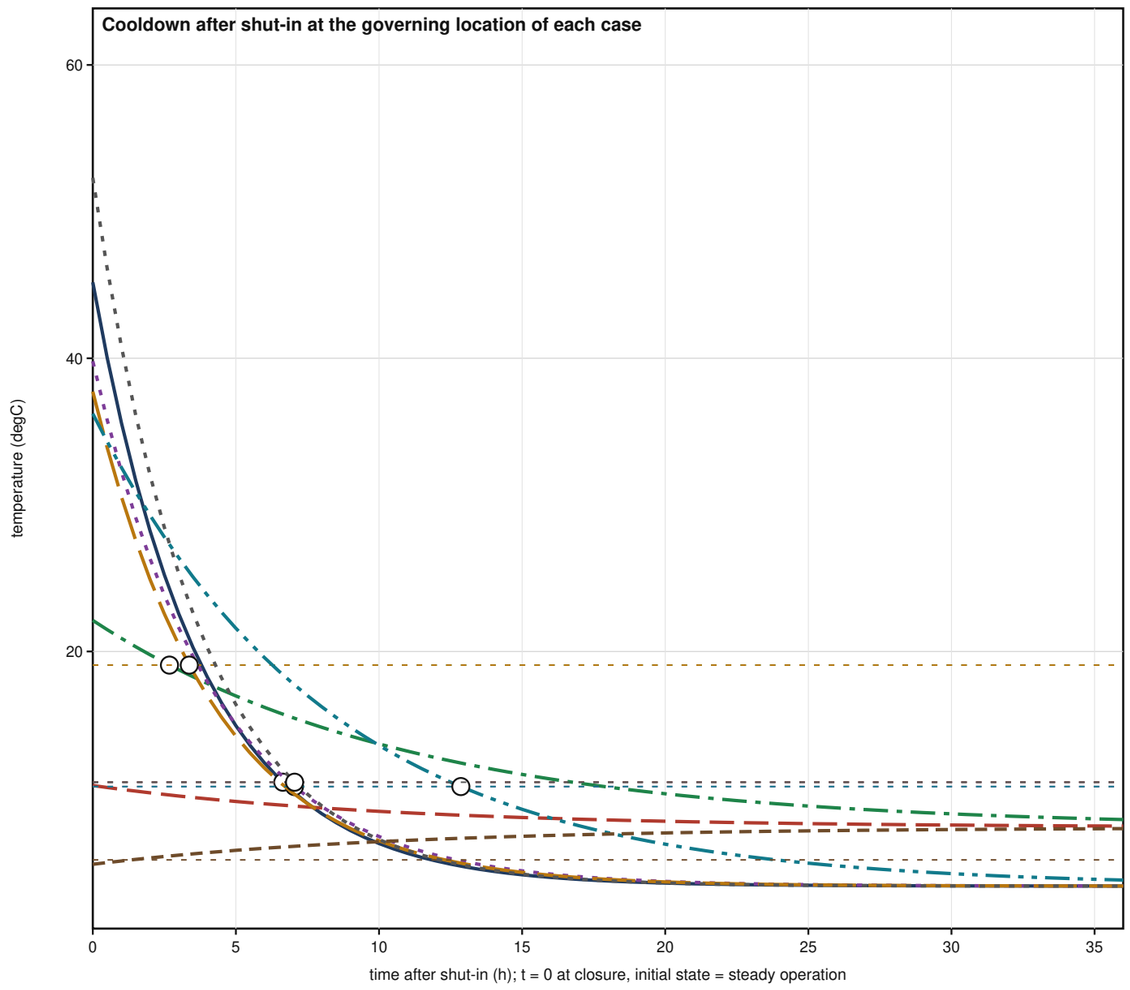
GENERAL NOTES

1. Values are the governing (worst) values over the whole route for each case; locations are on sheets 7 and 8.
2. Criteria are project inputs with a stated source (basis); none is presented as a code requirement. A criterion that cannot be assessed is shown as unavailable.
3. Threats that are open or unsupported have no result and are not shown as passed; see the report for reasons and hand-offs.
4. Cases: role design, scenario or mitigation as stated in the case matrix.
5. Issue purpose: DEMONSTRATION - SYNTHETIC DATA; readiness SCREENING (assigned by the toolkit gates, not by a person).
6. Run reference: dataset bd5858a7e40e53e449cc4021, model fa_solver steady network model, NeqSim neqsim-3.23.0-java25.jar.
7. Units: pressure bara, temperature degC, lengths m, rates kg/s, time h.

Rev	Date	Description
A	2026-10-25	First issue of the synthetic demonstration package

Company: NeqSim demonstration (no company)		Project: Synthetic tie-back (no real field)	
Title: **Synthetic Offshore Tie-back - Flow-Assurance Study (Demonstration) - Operating-limit summary by case**			
Drawing number: **FA-DEMO-001-DWG-09**		Revision: A	Status: **SCREENING**
Sheet: Sheet 9 of 10	Date: 2026-10-25	Scale: NOT TO SCALE	Format: A3 landscape
Drawn by: not assigned	Checked by: not assigned	Approved by: not assigned	

FA-DEMO-001-DWG-10 Transient and event sheet (cooldown, external hand-offs)



Event register (time origin: t = 0 at the stated event)

EVT-SHUTIN-01 | case BASE | shut_in
CALCULATED (screening, lumped cooldown)
governing time to the hydrate temperature: 6.6 h at KP 8.7 km
Unplanned shut-in from base operation; time to reach the hydrate equilibrium temperature at the settle-out pressure

EVT-SHUTIN-02 | case TURNDOWN | shut_in
CALCULATED (screening, lumped cooldown)
governing time to the hydrate temperature: 0.0 h at KP 25.7 km
Shut-in from turndown operation

EVT-SHUTIN-03 | case LATE | shut_in
CALCULATED (screening, lumped cooldown)
governing time to the hydrate temperature: 2.7 h at KP 26.1 km
Shut-in from late-life operation at the baseline MEG dose

EVT-RESTART-01 | case LATE | restart
EXTERNAL HAND-OFF - NOT SIMULATED - RESULT OPEN
Restart after 48 h shut-in with depressurisation, warm-up and liquid sweep-out through the riser
hand-off package: handoff/EVT-RESTART-01.json

EVT-PIG-01 | case BASE | pigging
EXTERNAL HAND-OFF - NOT SIMULATED - RESULT OPEN
Pig run from the pig launcher interface at PLEM-1 to the host pig receiver
hand-off package: handoff/EVT-PIG-01.json

Qualification basis of the calculated events

Lumped exponential cooldown (fluid + steel). Numerically verified against RK4 and the closed form;
NOT experimentally qualified. Result = time to the equilibrium temperature, not time to plug.
No simulated slug, pig or restart output is shown: those analyses are open (external simulator).

SCREENING - synthetic inputs; not for design or construction

LEGEND

- BASE - Base - plateau, early life
- TURNDOWN - Turndown - minimum stable rate
- LATE - Late life - high water, declining rate
- LATE-M1 - Late life - MEG dose raised to 30 wt % aqueous
- LATE-M2 - Late life - trunk insulation +30 mm, baseline MEG dose
- LATE-M3 - Late life - MEG 30 wt % and branch insulation 80 mm
- TURN-M4 - Turndown - MEG dose raised to 40 wt % aqueous
- ALT-B-ISOLATED - Alternate line-up - template B isolated
- Time to the hydrate equilibrium temperature (dotted line of the same style)

GENERAL NOTES

1. Initial condition is the steady operating temperature profile of the case; ambient is the segment ambient of the route file.
2. Settle-out pressure and the retained inhibitor mass fraction determine the criterion temperature of each case (basis: shut_in, assumption A-05).
3. Event chronology: shut-in events start at t = 0; external events (restart, pigging) carry their own time origin in the hand-off package.
4. A restart, pigging or slug-receiving result requires a separately qualified simulator; none is simulated or implied here.
5. Issue purpose: DEMONSTRATION - SYNTHETIC DATA; readiness SCREENING (assigned by the toolkit gates, not by a person).
6. Run reference: dataset bd5858a7e40e53e449cc4021, model fa_solver steady network model, NeqSim neqsim-3.23.0-java25.jar.
7. Units: pressure bara, temperature degC, lengths m, rates kg/s, time h.

Rev	Date	Description
A	2026-10-25	First issue of the synthetic demonstration package

Company: NeqSim demonstration (no company)		Project: Synthetic tie-back (no real field)	
Title: **Synthetic Offshore Tie-back - Flow-Assurance Study (Demonstration) - Transient and event sheet (cooldown, external hand-offs)**			
Drawing number: **FA-DEMO-001-DWG-10**		Revision: A	Status: **SCREENING**
Sheet: Sheet 10 of 10	Date: 2026-10-25	Scale: NOT TO SCALE	Format: A3 landscape
Drawn by: not assigned	Checked by: not assigned	Approved by: not assigned	